

Answer Key — Physics Grade 10

Sub-Strand 1.1: Pressure • Lesson 2: What Is Pressure? Measuring the Invisible Push — Shoes, Scales, and Force Per Unit Area

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
C	B	C	D	A	C	B	B

USE AFTER: PREDICT PHASE — Silent Force-Diagramming of the collapsed jerrycan

Q1. A sealed empty plastic jerrycan has had some air removed from inside and it is crumpling inwards. Which statement best describes how the air OUTSIDE is acting on it?

- A. It pushes only on the sides, because the top and bottom are flat
- B. It pushes only downwards on the top of the jerrycan
- C. It pushes inwards on every face: top, bottom and all four sides**
- D. It does not push at all; the plastic simply becomes weak

Correct answer: C. Atmospheric pressure acts on every exposed face of an object, so the outside air pushes inwards on all six faces. Students who draw arrows only on the top hold the 'downward only' misconception.

USE AFTER: OBSERVE PHASE — Part A: Pencil vs. Book Demonstration

Q2. A teacher presses a flat textbook onto her palm, then presses a sharp pencil tip onto her palm with about the same force. The pencil feels much more painful. What is the best explanation?

- A. The book has a smaller area, so it spreads the force less
- B. The force is about the same, but the pencil tip has a much smaller contact area, so the pressure is greater**
- C. The pencil is heavier than the book
- D. The pencil tip pushes with a much bigger force than the book

Correct answer: B. Pressure is force divided by area ($P = F/A$). With the same force, a smaller contact area gives a larger pressure.

USE AFTER: OBSERVE PHASE — Part B: Shoe Pressure Investigation

Q3. A student has a mass of 50 kg ($g = 10 \text{ N/kg}$). She stands on one shoe whose sole has a contact area of 200 cm^2 . Taking her weight as shared equally between both feet, what pressure does one shoe exert on the floor? ($1 \text{ m}^2 = 10,000 \text{ cm}^2$)

- A. 1.25 Pa
- B. 25,000 Pa
- C. 12,500 Pa**
- D. 125,000 Pa

Correct answer: C. Weight = $50 \times 10 = 500 \text{ N}$, so one foot carries 250 N. Area = $200 \div 10,000 = 0.02 \text{ m}^2$. $P = 250 \div 0.02 = 12,500 \text{ Pa}$.

USE AFTER: EXPLAIN PHASE — Formal definition and note frame ($P = F/A$, unit Pascal)

Q4. Which choice correctly gives the definition of pressure and its SI unit?

- A. Pressure = area $\div$ force; unit is the Newton (N)
- B. Pressure = mass $\div$ area; unit is the kilogram (kg)
- C. Pressure = force $\times$ area; unit is the square metre (m^2)
- D. Pressure = force $\div$ area; unit is the Pascal (Pa), which is N/m^2**

Correct answer: D. Pressure is force per unit area, $P = F/A$, measured in Pascals where $1 \text{ Pa} = 1 \text{ N/m}^2$. Inverting the formula or using mass instead of force are common errors.

USE AFTER: EXPLAIN PHASE — Extension Problem 1: maize bag on a wooden pallet

Q5. A maize bag of mass 90 kg ($g = 10 \text{ N/kg}$) is placed on a wooden pallet in a warehouse in Eldoret. The pallet has 4 legs, each with a base area of 25 cm^2 . The weight is shared equally by the legs. What pressure does each leg exert on the floor? ($1 \text{ m}^2 = 10,000 \text{ cm}^2$)

- A. 90,000 Pa**
- B. 9,000 Pa
- C. 900,000 Pa
- D. 360,000 Pa

Correct answer: A. Weight = $90 \times 10 = 900 \text{ N}$; each leg carries $900 \div 4 = 225 \text{ N}$. Area = $25 \div 10,000 = 0.0025 \text{ m}^2$. $P = 225 \div 0.0025 = 90,000 \text{ Pa}$.

USE AFTER: EXPLAIN PHASE — Extension Problem 2: rungu and crutch comparison

Q6. A Maasai elder leans on a rungu whose handle tip has an area of 2 cm^2 , with a force of 150 N. A nurse at Kenyatta National Hospital leans on a flat crutch base of area 50 cm^2 with the same 150 N force. How do the pressures on the floor compare?

- A. The rungu exerts 48 times more pressure than the crutch
- B. The pressures are equal, because both forces are 150 N
- C. The rungu exerts 25 times more pressure than the crutch**
- D. The crutch exerts 25 times more pressure than the rungu

Correct answer: C. Rungu: $150 \div 0.0002 = 750,000 \text{ Pa}$. Crutch: $150 \div 0.005 = 30,000 \text{ Pa}$. The ratio is $750,000 \div 30,000 = 25$, so the crutch is safer for a soft floor.

USE AFTER: DQB PHASE — Two-colour DQB cards: 'What do I now THINK is causing the jerrycan to crush?'

Q7. Which statement best describes atmospheric pressure?

- A. It is a force that acts only downwards on the top of objects
- B. It is the weight of the column of air above a surface, and it acts on every exposed face of an object**
- C. It is the force of moving wind, and it acts only in the direction the wind blows
- D. It is the pressure of the air trapped inside a container only

Correct answer: B. Atmospheric pressure comes from the weight of the air column above a surface and it pushes on all exposed faces, not just the top.

USE AFTER: MODEL BUILDING PHASE — Revised jerrycan model: sentence frame 'Before the air was removed... After air was removed...'

Q8. An empty sealed jerrycan with normal air inside and outside stays intact. After some air is pumped out of it, it collapses. Which explanation is correct?

- A. Before, there was no air pushing on the jerrycan at all
- B. Before, the air pressure inside pushing outwards balanced the air pressure outside pushing inwards; after, the pressure inside was much less, so the outside pressure crushed it**
- C. The air outside became heavier after the air inside was removed
- D. The plastic was already weak and only needed a small push from the pump

Correct answer: B. With equal pressure inside and outside the inward and outward pushes balance. Removing air lowers the inside pressure, so the unbalanced outside pressure collapses the jerrycan.